Metals

Structure of metals: A regular pattern (lattice) made up of positive ions packed together, surrounded with a sea of a mobile or delocalized electrons

Metallic bond: The electrostatic attraction between the positive ions in a giant metallic lattice and a sea of delocalized electrons.

Physical properties of metals:

* High melting and boiling points: All solid except mercury
Because lots of energy is required to break the strong forces of attraction between the metal ion and the delocalized electrons.
* Good conductors of electricity & heat: Because free electrons take in heat energy, which makes them move faster. And they quickly transfer the heat through the metal
* Malleable → Can be hammered into shapes without breaking
* Ductile → Can be drawn to wires.
Because the positive ions are arranged in regular layers, that can slide over each other without breaking. And because electrons are free to move too.
* Strong
* Sonorous → Make ringing sound when struck.
* Shiney when polished
* High density.

Chemical properties of metals:

* They react with oxygen to form oxides
* Most metal oxides are basic oxides, so they're able to neutralize acids ⟶ forming salts + water
* Most metals react with dilute acids to form salts and hydrogen.
* Metals form positive ions when they react.
* Charge of ion is same as group number, except for transition metals, because they have varying valencies.

→ Group (VII): Halogens: Are the most reactive non-metals

→ Group (VIII): Noble gases: Un-reactive because their atoms have full outer shell.

Physical properties of non-metals:

1- low melting & boiling points, due to the weak forces of attraction between molecules, or the weak intermolecular forces.

2- Brittle, not malleable or ductile, they break to small pieces when hammered

3- Solid non-metals do not conduct electricity or heat

4- Solid non-metals look dull

5- Solid non-metals have low density.

Chemical properties of non-metals

1- They react with oxygen to form acidic oxides.

2- When they react with metals, they form negative ions. forming ionic compounds

3- When reacting with other non-metals, they form covalent compounds.

Carbon allotropes: - Diamond and - Graphite have giant molecular structure

* Diamond is very hard, and have high melting & boiling points
* Graphite can conduct electricity.

The reactivity of a metal is the ability to form a positive ion, the strong drive to become a compound.

→ The more easily its atoms give up electrons, the more reactive a metal will be.

* Reaction with oxygen in Air :

Metal + oxygen ⟶ Metal oxide

* They are basic oxides, some of them are insoluble in water, some of them are soluble.

→ Most reactive metals react with oxygen with a bright flame and producing white ashes and their oxides are soluble.

→ Moderately reactive metals like aluminum and zinc react with white powdered ashes but their oxides are insoluble.

Potassium
Sodium
Calcium
Magnesium
Aluminum
Carbon
Zinc
Iron
Tin
lead
Hydrogen
Copper
Silver
Gold
Platinum

React with cold Water
Metal hydroxide + H_2

Reacts with steam
Metal oxide + H_2

Seems to unreactive but it is reactive if the aluminum oxide layer was removed

do not with water nor steam

→ Iron and copper react very slowly with oxygen. The result iron oxygen is rust, which is reddish brown iron (III) oxide. When a copper lump reacts with oxygen, a layer of black copper oxide forms on it. Oxides of iron and copper are insoluble in water

→ Gold, silver and platinum don't react with water

→ Aluminum reacts with oxygen, a layer of aluminum oxide adheres and covers the aluminum. At this point no further reaction can take place.

Sodium : Burns vigorously with a yellow flame : Na_2O
White solid

Calcium : Burns vigorously on strong heating, with an orange-red flame : CaO
White solid.

Magnesium, Burns readily with a brilliant white light : MgO
White solid

Iron : Burns readily with bright sparks when iron fillings are used : FeO
Black solid

Copper : Reacts slowly at the surface, with a green-blue flame, and a black coat form : CuO
Black solid

* Reactions of metals with Water and steam

Metal + Water ⟶ Metal hydroxide + Hydrogen

* Potassium and sodium react vigorously with cold water and may catch on fire. Hydrogen gas being produced accumulates it may ignite and cause an explosion.
* Calcium reacts readily but not violently with cold water to form hydrogen and calcium hydroxide solution, the alkali lime water
* Magnesium, aluminum, zinc and iron are less reactive. They react with steam forming metal oxide and hydrogen. Magnesium, aluminum will react vigorously with steam while zinc and iron react slowly.

Metal + Steam ⟶ Metal oxide + Hydrogen.

* Magnesium reacts very slowly with cold water to form hydrogen and magnesium hydroxide, but it reacts vigorously and brightly with steam to form a white powder. Magnesium oxide

* Copper, silver, gold, and platinum don't react with water or steam

→ Metals with water or steam:
* Single displacement reaction where hydrogen is replaced with a more reactive metal.
* Redox reaction
* Exothermic reaction

* The reaction of metals with hydrochloric acid: A metal and an acid are used to prepare soluble salts, and this method is only suitable preparing salts of moderately reactive metals:

- Magnezium
- Aluminum
- Zinc
- Iron
- tin

MAZIT

* Metals more reactive than magnesium will react very violently with acids is very dangerous. So it is not safe to add sodium or potassium to acid in the lab, because reactions are explosively fast

$$\underset{(s)}{\text{Metal}} + \underset{(aq)}{\text{Acid}} \longrightarrow \underset{(aq)}{\text{Salt}} + \underset{(g)}{\text{Hydrogen}}$$

→ It is dangerous to react alkali metals because they react explosively fast with acids.

→ When a metal does react with water or hydrochloric acid, it drives and takes it's place. So the metal is more reactiv than hydrogen, it has a stronger drive to exist as a compound.

* The more reactive metal has a greater tendency to form a metal ion by losing electrons than a less reactive metal does. So if a more reactive metal is heated with the oxide of a less reactive metal, it will remove the oxygen from it.

→ A metal will reduce the oxide of a less reactive. The reduction always gives out heat, it is exothermic. The metal itself is oxidized.

When aluminum reacts with iron(III) oxide. Aluminum is oxidized and the iron ion is reduced. This is a thermite reaction, since large amounts of heat are given out, and the iron in formed in molten state. This is used to weld damaged railway lines

* If a metal oxide reacts with carbon, and the metal is more reactive than carbon, then no reaction will happen. However, if the metal is less reactive, carbon will reduce the metal, and become oxidized

*When zinc is added in excess to copper (II) nitrate solution, zinc will get coated with copper, and the solution turns colorless

$$Zn_{(s)} + Cu(NO_3)_{2\,(aq)} \rightarrow Zn(NO_3)_{2\,(aq)} + Cu_{(s)}$$

Observations

* Blue color of the solution fades
* Zinc is coated with red-brown solid (Copper)

Half equations:

$$Zn_{(s)} \longrightarrow Zn^{2+}_{(aq)} + 2e^-$$

$$Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$$

$$Zn_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$$

→ A more reactive metal will displace a less reactive metal from a solution

$$Cu_{(s)} + 2AgNO_{3\,(aq)} \longrightarrow Cu(NO_3)_{2\,(aq)} + 2Ag_{(s)}$$

Observations:

- Colorless solution turns blue
- Copper gets coated with a silvery solid

$$Cu_{(s)} \longrightarrow Cu^{2+}_{(aq)} + 2e^-$$

$$2Ag^+_{(aq)} + 2e^- \longrightarrow 2Ag_{(s)}$$

$$Cu_{(s)} + 2Ag^+_{(aq)} \longrightarrow Cu^{2+}_{(aq)} + Ag_{(s)}$$

* Metal atoms act as reducing agents, so they get oxidized and lose electrons. The reducing power decreases down from the most reactive to the least reactive metal.

→ Atoms reduce all the ions below them.

* Metal ions act as oxidizing agents, so they get reduced and gain electrons. The oxidizing power increases down the group.

→ Ions oxidize all the atoms below them

* The more reactive metal atoms reduce the less reactive metal ions by giving them electrons.
* The less reactive metal ions oxidize the more reactive metal atoms by accepting their electrons.

Aluminum ore: Bauxite : Contains 50-65% Aluminum oxide, as well as other impurities.

Steps of extraction of aluminum from bauxite:

1. Purify the ore by reacting the crushed ore with sodium hydroxide (NaOH), aluminum oxide with the NaOH because it is amphoteric oxide, and dissolves to form sodium aluminate ($NaAlO_2$)

$$Al_2O_{3(s)} + 2NaOH_{(aq)} \longrightarrow 2NaAlO_{2(aq)} + H_2O_{(l)}$$

* The impurities are insoluble in sodium hydroxide, so we get rid of them by filtration.
* Sodium aluminate is heated to make pure aluminum oxide, Al_2O_3

2. Electrolyse aluminum oxide. Al_2O_3 must be molten, but the melting point is very high: 2000°C, which is too costly because it needs so much energy. So aluminum oxide is not melted, it's dissolved in molten cryolite. Which is a good solvent for aluminum oxide, and the solution is a better conductor than Al_2O_3

Reasons to dissolve aluminum oxide in molten cryolite:

1- Decreases the melting point of aluminum oxide to 900°C

2- Saves a lot of energy

3- Improves the electrical conductivity of the electrolyte

4- Acts as a solvent

* At the cathode: Aluminum is reduced to aluminum metal, and sinks to the bottom of the cell in molten form because it is very dense.

$$Al^{3+}_{(aq)} + 3e^- \longrightarrow Al_{(l)}$$

* At the anode: Oxide ions are oxidized to oxygen molecules. Oxygen gas reacts with hot carbon electrodes, and form carbon dioxide, which is why the anode electrode should be replaced

$$2Al_2O_{3\,(l)} \longrightarrow 4Al_{(l)} + 3O_{2\,(g)}$$

Uses of aluminum,

→ Construction & spacecraft bodies because it is very strong and very light. It is also resistant to corrosion.

→ Food containers because it is resistant to corrision; the thin layer of aluminum oxide prevents further reactions.

→ Overhead power cables because it conducts electricity, is very light, malleable & ductile. It is strengthened by steel core.

* It is not strong as iron, but makes strong alloys
* Conducts electricity, but not as good as copper

Iron ore: Hematite: contains 60% iron (III) oxide, and other impurities like silicon (IV) oxide.

→ Extracted using a blast furnace

Charge: Contains:

1. Iron ore
2. limestone
3. Coke: Cole and almost pure carbon

Steps for the extraction of iron:

1. Blowing hot air at the bottom of the furnace; so the coke will burn in oxygen from hot air to produce carbon dioxide, at 1400°C

* Combustion reaction
* Highly exothermic

$$C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)}$$

2. Carbon dioxide produced goes up the furnace and reacts with more coke to produce carbon monoxide. The carbon reduces the carbon dioxide into carbon monoxide at 800°C

$$CO_{2(g)} + C_{(s)} \longrightarrow 2CO$$

* CO is the main reducing agent
* Endothermic reaction as temperature drops from 1400 to 800°

* Carbon monoxide is also formed from the incomplete combustion of coke at the top of the furnace.

3. Carbon monoxide rises further up the furnace where it meets iron oxide, and starts reducing it to produce iron and carbon dioxide at 400°C

$$Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$$

Limestone is used to remove impurities:

① Heat makes the limestone to decompose into calcium oxide and carbon dioxide

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$

② Calcium oxide is a base. It reacts with impurities of hematite such as silicon oxide which is acidic, forming Calcium silicate, which is called slag

$$CaO + SiO_2 \longrightarrow CaSiO_3$$

* Molten iron & slag trickel down and settle at the bottom of the furnace. The iron is denser, so it settles below slag, so the slag forms a protective layer above the iron to prevent further oxidation.

* The waste gases, mainly oxides of carbon & nitrogen escape from the top of the surface. They are used to heat incoming air, to help reduce the energy costs of the process.

* Slag is also a waste material that is used by builders and road makers for foundation.

* Iron that comes out of the blast furnace is called pig iron, it is impure, as it contains 5% carbon which makes it brittle, along with sand and other impurities.

* Pig iron is run into moulds to make cast iron, which is hard but still brittle because of the high carbon content. Which makes it only usable for bottled gas canistres and drain convers.

	% Carbon content	Properties	Uses
* Mild steel	< 0.25% carbon	Easily worked & not brittle	Used in car bodies, chains, large structures, machinery
* Stainless steel	Iron + nickel + Chromium + Carbon & other elements	Strong resistant to corrosion and looks attractive	Cutlery, cooking utensils, surgical equipment, kitchen sinks, chemical plants & construction

Why is recycling important?

1- Conserves raw materials

2- Uses less energy to recycle something than the energy needed to make it from new materials.

3- Avoids the need to bury substances in landfill sites, which avoids pollution

Why is iron & steel the most recycled metals in the world?

1- It costs far less to recycle them than to mine new ores and extract the iron. It's also better for the environment.

2- It is easy to separate them from scrap using a magnet.

3- It is easy to recycle them by just adding them to oxygen furnace.

	Used for :	Properties :
* Aluminum :	- Overhead electricity cables	- It is a good conductor of electricity, ductile, and resists corrosion, cheaper & much lighter than copper.
	- Cooking foil & food cartons	- Non-toxic, resists corrosion, can be rolled into thin sheets.
	- Drink cans	- Light, non-toxic, resist corrosion.
	- Coating CDs & DVDs	- Can be deposited as a thin film, shiny surfaces reflect laser beam.
	- Aircraft	- Low density and resists corrosion
* Copper :	- Electrical wiring	- One of the best conductors of electricity, ductile
	- Saucepans and saucepan bases	- Malleable, conducts heat well, unreactive, tough.
	- Pipes for water supply inside homes	- Malleable, doesn't corrode easily, non-toxic
* Zinc :	- Protecting steel from rusting	- Offers sacrificial protection to the iron in steel
	- Coating or galvanising iron and steel	- Resists corrosion, also offers sacrificial protection if coating cracks
	- Torch batteries	- Gives a current when connected to a carbon pole, packed into a paste of electrolyte

Alloy: A mixture of two or more metals, or one or more metals with a non-metal

Why does an alloy have different properties?

The atoms in a pure metal are arranged in regular layers, so when a force is applied, the layers slide over each other easily without breaking. Which makes them malleable and ductile. However, in an alloy, the different sized metal atoms make the arrangement of the lattice less regular, this stops the layers of metal atoms from sliding over each other easily when a force is applied. Which is why an alloy is harder than a pure metal.

* Brass: Copper and zinc : Harder than copper easy to shape, looks attractive, does not corrode : Musical instruments, ornaments, door knobs

* Stainless Steel : Iron, chromium, nickel, carbon : Strong, resists corrosion, looks attractive : Cutlery, cooking utensils, construction surgical equipment, in factories equipment

* Solder: lead + tin: lower melting point : Connecting electrical wiring

Rust: A red-brown powder consisting mainly of hydrated iron (III) oxide.

* Water & oxygen are essential for iron to rust, so if one of them is not present, rusting will not occur. The presence of seawater increases the rate of corrosion because of the salt.

→ Iron gets oxidized by air and water to form hydrated iron (III) oxide

$$4Fe_{(s)} + 3O_{2(g)} + 4H_2O_{(l)} \longrightarrow 2Fe_2O_3 . 2H_2O_{(s)}$$

How does aluminum not corrode in the damaging way that iron does even though it is more reactive?

A single layer of aluminum oxide forms which strikes strongly to the metal surface and protects the surface of the metal. This layer seals the surface and protects it from further attacks.

However, when iron corrodes, it forms flakes, not a single layer, so the attack on the metal continues as the flakes come off.

→ Chromium is like aluminum, protected with an oxide layer

Ways to prevent rusting:

1- Physical barriers: Covering the iron & steel to keep the air and water.

* Painting: Used for objects in size of ships and bridges. It can only protect the metal if the paint layer is unscratched.

* Oiling & greasing: It prevents the moving parts of machinery from coming into contact with air and moisture.

* Plastic coating: Used to form a protective layer on items such as refrigerators and garden chairs.

* Coating with another metal: Iron and steel will be electroplated with a layer of zinc, chromium, or tin. Tin is used because it is unreactive and non-toxic, if this layer is broken, the steel beneath will rust. Steel is protected by zinc in car bodies.

* Galvanising: An object will be coated by a more reactive metal, zinc. It will work even if the zinc is badly scratched.

How does the iron not rust even if the zinc layer is scratched?

If the zinc layer is scratched, a simple cell is set up, with zinc as the anode. Zinc is more reactive than iron, so it is loses its electrons to the iron. By giving up electrons, zinc prevents the iron from giving up electrons and becoming oxidized; so the zinc corrodes rather than iron. As electrons will always flow from zinc to iron, iron is always reduced.

$$Zn \longrightarrow Zn^{2+} + 2e^-$$

$$Fe^{2+} + 2e^- \longrightarrow Fe$$

$$Zn + Fe^{2+} \longrightarrow Zn^{2+} + Fe$$

2- Sacrificial protection: letting another metal corrode instead.

* Zinc and magnesium blocks are attached to oil rigs and the hulls of ships.
* Underground gas and water pipes are connected to blocks of magnesium.

→ In all cases, an electrochemical cell is set up, the metal block loses electrons in preference to iron because it is more reactive than iron, and so prevents the iron from getting oxidized to form iron (III) oxide

* As the more reactive metal corrodes, it loses electrons to iron and so protects it, by keeping the iron reduced.

→ If the zinc coat on galvanized iron gets damaged, the zinc will still protect the iron. So galvanizing provides a barrier and sacrificial protection.

3- Electrolytic protection / Cathodic protection: large, static steel structures are protected using this method.

* An electrolytic cell is set up where the iron or steel is at the cathode of the cell. An inert electrode and a power supply are needed. The iron at the cathode will always remain reduced, and at the anode, oxidation will happen.

Differences between Sacrificial protection and cathodic protection:

* Sacrificial cell does not need electricity
* Cathodic is electrolysis, so needs electricity.
* Sacrificial protection needs a more reactive metal in contact with iron.
* Cathodic only need an inert electrode as an anode